



MID SEMESTER EXAMINATION, AUTUMN-2024
School of Applied Sciences, KIIT-DU

B. Tech.
1st Semester (2024AB)
Autumn 2024 (SAS)

Course: Optimization Technique

Code: MA10003

Full Marks: 20

Time: 90 minutes

Answer any FOUR QUESTIONS including question No. 1 which is compulsory.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

All parts of a question should be answered at one place only.

Q.1		Marks	CO
a)	What are the basic components in the formulation of a LPP?	1	CO1
b)	Write the standard form of a LPP.	1	CO1
c)	Draw the feasible region for the following constraints: $\begin{aligned}x_1 + x_2 &\leq 3 \\ -x_1 + 2x_2 &\leq 4 \\ x_1, x_2 &\geq 0.\end{aligned}$	1	CO2
d)	Find the maximum and minimum value of $Z = 2x_1 + 3x_2$, where the constraints are as mentioned in Q.1(c).	1	CO2
e)	Define the feasible solution of a LPP.	1	CO2

Q.2		Marks	CO
	The University science lab is conducting two experiments: Experiment A and Experiment B . Each run of Experiment A requires 4 hours of equipment time and 3 hours of lab technician time, while each run of Experiment B requires 2 hours of equipment time and 1 hour of lab technician time. The lab has a total of 80 hours of equipment time and 60 hours of technician time available each week. The university earns \$100 for each run of Experiment A and \$70 for each run of Experiment B . Formulate the LPP to maximize the weekly earnings from the experiments.	5	CO1

Q.3		Marks	CO
	Solve the following LPP by using graphical method: $\begin{aligned}\text{Minimize } Z &= 3x_1 + 4x_2 \\ \text{Subject to } & \\ &2x_1 + x_2 \geq 8 \\ &x_1 + 2x_2 \geq 10 \\ &x_1, x_2 \geq 0.\end{aligned}$	5	CO2

Q.4		Marks	CO
	Use graphical method to solve the following LPP: $\begin{aligned}\text{Maximize } Z &= 6x_1 + 4x_2 \\ \text{Subject to } & \\ &-2x_1 + x_2 \leq 2 \\ &x_1 - x_2 \leq 2 \\ &3x_1 + 2x_2 \leq 9 \\ &x_1, x_2 \geq 0.\end{aligned}$	5	CO2

Q.5		Marks	CO
	An individual wishes to invest \$5000 over the next year in two types of investment: Investment A yields 5% and investment B yields 8%. Market research recommends an allocation of at least 25% in A and at most 50% in B. Moreover, investment in A should be at least half the investment in B. How should the fund be allocated to the two investments? Formulate the above problem as a LPP.	5	CO1